

PhD Math Camp

Linear Algebra & Intro to Multivariate Calculus

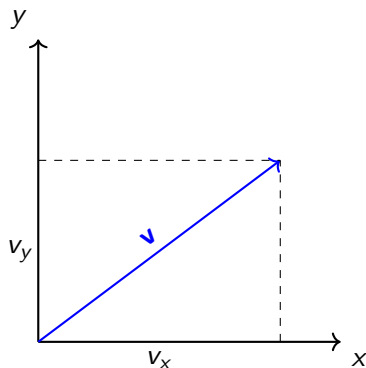
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Vectors

Introduction to Vectors

- A vector is a quantity that has both magnitude and direction.
- Represented as an arrow in space.
- Commonly expressed in component form, e.g., $\mathbf{v} = \begin{pmatrix} v_x \\ v_y \end{pmatrix}$ in 2D.

Vectors - Geometric Interpretation



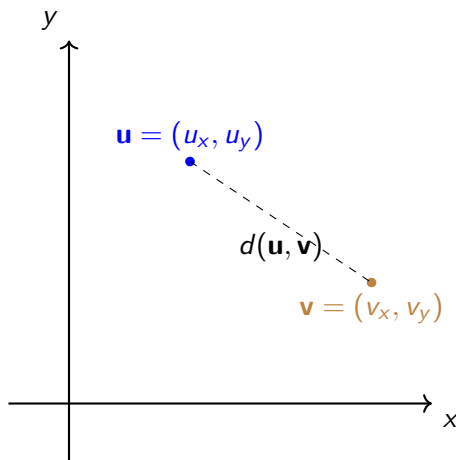
- $\mathbf{v}$ can be broken down into horizontal (v_x) and vertical (v_y) components.
- These components form the sides of a right triangle.

Spatial Interpretation of Vectors

- Vectors can be interpreted not just as arrows, but as points in a given space.
- A vector $\mathbf{v} = \begin{pmatrix} v_x \\ v_y \end{pmatrix}$ in 2D can be seen as the point (v_x, v_y) in the plane.
- This interpretation allows us to talk about the **distance** between points (vectors) in space.
- The distance between two vectors (points) $\mathbf{u}$ and $\mathbf{v}$ is given by the Euclidean distance formula:

$$d(\mathbf{u}, \mathbf{v}) = \sqrt{(u_x - v_x)^2 + (u_y - v_y)^2}$$

Points as Vectors in Space



- Vectors $\mathbf{u}$ and $\mathbf{v}$ can be viewed as points in the 2D space.
- The Euclidean distance $d(\mathbf{u}, \mathbf{v})$ represents how "far apart" these points (vectors) are.

Example: Multivariate Data Representation

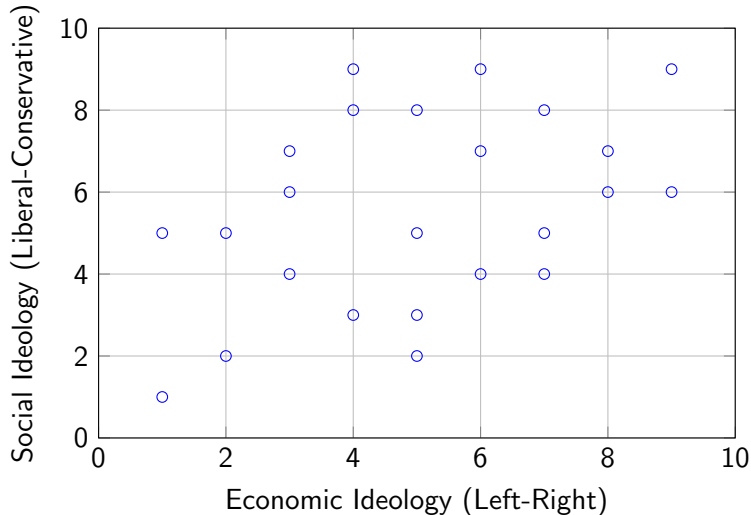
- Consider a dataset with three variables: height, weight, and age for different individuals.

- Each individual can be represented as a vector in $\mathbb{R}^3$, e.g.,

$$\mathbf{v}_1 = \begin{pmatrix} 180 \\ 75 \\ 25 \end{pmatrix} \text{ (height in cm, weight in kg, age in years).}$$

- This vector representation allows for the analysis of patterns, relationships, and clusters within the data.

Scatterplot: Economic vs. Social Ideology



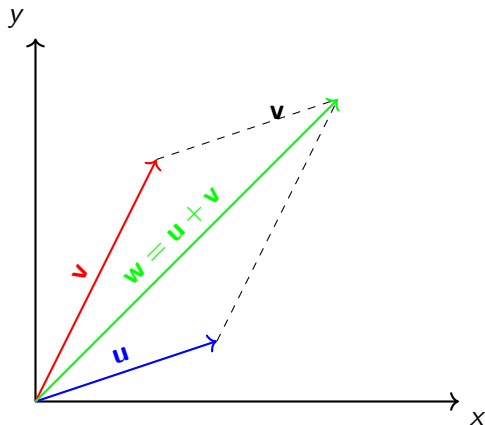
- **Figure:** All these points can be thought of as vectors.

Vector Operations

Vector Addition and Subtract

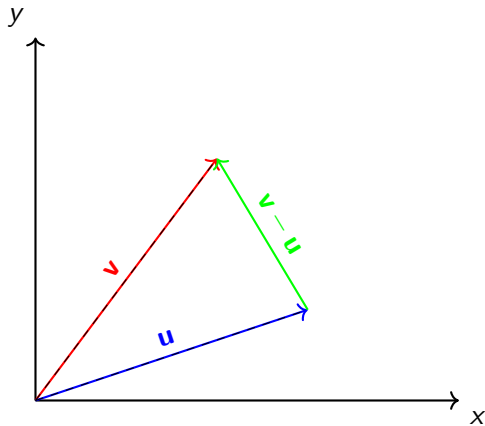
- To add two vectors, $\mathbf{u}$ and $\mathbf{v}$, you simply add their corresponding components.
- Algebraically: $\mathbf{u} + \mathbf{v} = \begin{pmatrix} u_x + v_x \\ u_y + v_y \end{pmatrix}$
- To subtract two vectors, you simply subtract their corresponding components.
- Algebraically: $\mathbf{u} - \mathbf{v} = \begin{pmatrix} u_x - v_x \\ u_y - v_y \end{pmatrix}$

Graphical Example of Vector Addition



- **Figure:** A graphical example of the vector addition $u + v$.

Graphical Example of Vector Subtraction

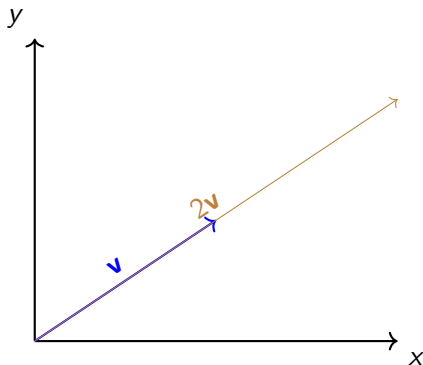


- **Figure:** A graphical example of vector subtraction $\mathbf{v} - \mathbf{u}$.

Scalar Multiplication

- Multiplying a vector by a scalar changes its magnitude but not its direction.
- To perform the operation, multiply the scalar k by each component of the vector.
- Algebraically: $k\mathbf{v} = \begin{pmatrix} kv_x \\ kv_y \end{pmatrix}$

Scalar Multiplication - Geometric Interpretation



- Scaling by a factor of 2 doubles the length of the vector, maintaining the same direction.
- If the scalar were negative, the direction would reverse.

Linear Combination of Two Vectors

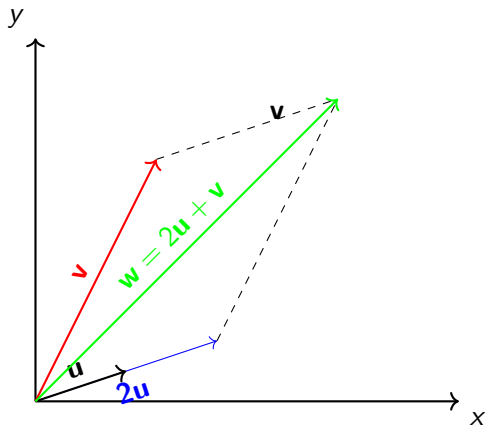
- Let $\mathbf{u} = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$ be two vectors in $\mathbb{R}^2$.
- A **linear combination** of $\mathbf{u}$ and $\mathbf{v}$ is any vector $\mathbf{w}$ that can be written as:

$$\mathbf{w} = c_1\mathbf{u} + c_2\mathbf{v} = c_1 \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} + c_2 \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$$

where c_1 and c_2 are scalar coefficients.

- This means that $\mathbf{w}$ is formed by scaling $\mathbf{u}$ and $\mathbf{v}$ by c_1 and c_2 respectively, and then adding the resulting vectors.
- The space of all possible linear combinations of $\mathbf{u}$ and $\mathbf{v}$ forms a plane (if the vectors are not collinear).

Graphical Example of a Linear Combination



- **Figure:** A graphical example of the linear combination $w = 2u + v$.

Why Create Linear Combinations of Vectors?

- **Question:** Why would we want to create linear combinations with vectors, especially if we interpret vectors as points in space or data points?
- **Intuition:** Linear combinations allow us to combine information from multiple data points (or vectors) to generate new insights.
- **Example:** One common use is to calculate the average of data points.

Example: Averaging Data Points Using Linear Combinations

- **Example:**

- Consider three individuals, each with data on two variables: economic ideology and social ideology.
- Let the data points (vectors) be:

$$\mathbf{v}_1 = \begin{pmatrix} 4 \\ 3 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 6 \\ 5 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 8 \\ 7 \end{pmatrix}$$

- A linear combination that averages these vectors is:

$$\mathbf{v}_{\text{avg}} = \frac{1}{3}\mathbf{v}_1 + \frac{1}{3}\mathbf{v}_2 + \frac{1}{3}\mathbf{v}_3 = \frac{1}{3} \begin{pmatrix} 4 \\ 3 \end{pmatrix} + \frac{1}{3} \begin{pmatrix} 6 \\ 5 \end{pmatrix} + \frac{1}{3} \begin{pmatrix} 8 \\ 7 \end{pmatrix} = \begin{pmatrix} 6 \\ 5 \end{pmatrix}$$

- **Conclusion:** The resulting vector $\mathbf{v}_{\text{avg}}$ represents the average economic and social ideology for the sample.

Linear Combinations in 3D

- Interactive Demo

Dot Product of Vectors

- The dot product of two vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^2$ is a scalar. Algebraically: $\mathbf{u} \cdot \mathbf{v} = u_x v_x + u_y v_y$.
- If $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^N$, then:

$$\mathbf{u} \cdot \mathbf{v} = \begin{pmatrix} u_1 \\ \vdots \\ u_N \end{pmatrix} \cdot \begin{pmatrix} v_1 \\ \vdots \\ v_N \end{pmatrix} = \sum_{i=1}^N u_i v_i$$

- Geometrically, it represents the product of the magnitudes of the two vectors and the cosine of the angle between them.
- **Example:** For $\mathbf{u} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$, the dot product is $3 \times 2 + 4 \times 1 = 10$.
- The dot product can be used to determine the angle between vectors and to check if they are perpendicular.

Norm of a Vector

- The norm (or magnitude) of a vector $\mathbf{v} = \begin{pmatrix} v_x \\ v_y \end{pmatrix}$ in $\mathbb{R}^2$ is given by $\|\mathbf{v}\| = \sqrt{v_x^2 + v_y^2}$, and it represents the *length* of the vector in Euclidean space.
- **Example:** For $\mathbf{v} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$, the norm is $\sqrt{3^2 + 4^2} = 5$.
- If $\mathbf{v}$ in $\mathbb{R}^N$, then:

$$\|\mathbf{u}\| = \left\| \begin{pmatrix} u_1 \\ \vdots \\ u_N \end{pmatrix} \right\| = \sqrt{\mathbf{u} \cdot \mathbf{u}} = \sqrt{\sum_{i=1}^N u_i^2}$$

- Important for normalizing vectors (making the length 1) which is often required in statistical analysis when comparing different data sets.

Distance Between Two Vectors

- The distance between the tips of two vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ is a key concept in vector mathematics.
- **Definition:** The distance is computed as the norm of the difference between the two vectors.
- **Formula:**

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| = \sqrt{\sum_{i=1}^N (u_i - v_i)^2}$$

- This formula represents the Euclidean distance, which is the straight-line distance between the two points in space represented by the vectors.

Example of Distance Calculation

- **Example:**

- Consider the vectors $\mathbf{u} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.
- Step 1: Compute the difference between the vectors:

$$\mathbf{u} - \mathbf{v} = \begin{pmatrix} 3 - 1 \\ 4 - 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

- Step 2: Calculate the norm of the difference:

$$d(\mathbf{u}, \mathbf{v}) = \left\| \begin{pmatrix} 2 \\ 3 \end{pmatrix} \right\| = \sqrt{2^2 + 3^2} = \sqrt{4 + 9} = \sqrt{13}$$

- The result, $\sqrt{13}$, is the Euclidean distance between the points represented by the vectors $\mathbf{u}$ and $\mathbf{v}$.

Orthogonality

Orthogonality and Projections

- Two vectors are orthogonal if their dot product is zero:
 $\mathbf{u} \cdot \mathbf{v} = 0$.
- The projection of vector $\mathbf{u}$ onto $\mathbf{v}$ is given by a new vector

$$\text{proj}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}} \mathbf{v}$$

- **Example:** Projecting $\mathbf{u} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ onto $\mathbf{v} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$ results in $\begin{pmatrix} 3 \\ 0 \end{pmatrix}$.
- In the next slides we will see why orthogonality implies a dot product equal to zero.

Norm and Angles I

1. Recall that the distance between point $\mathbf{v}$ and the origin is called the norm, a non-negative number. Pythagorean theorem implies that

$$\text{norm}(\mathbf{v}) = \sqrt{v_x^2 + v_y^2}$$

2. Let $\theta_{\mathbf{v}}$ be the angle between $\mathbf{v}$ and the x-axis.
3. From trigonometry we know that

$$\cos \theta_{\mathbf{v}} = \frac{v_x}{\sqrt{v_x^2 + v_y^2}} = \frac{v_x}{\text{norm}(\mathbf{v})}$$

$$\sin \theta_{\mathbf{v}} = \frac{v_y}{\sqrt{v_x^2 + v_y^2}} = \frac{v_y}{\text{norm}(\mathbf{v})}$$

Norms and Angles II

1. Suppose we have another column vector

$$\mathbf{u} = \begin{pmatrix} u_x \\ u_y \end{pmatrix}$$

In the xy -plane, now there are two points: one is for $\mathbf{v}$, and the other for $\mathbf{u}$.

2. Let $\theta_{\mathbf{u}}$ be the angle between $\mathbf{u}$ and the x -axis, and $\theta_{\mathbf{vu}}$ be the angle between $\mathbf{v}$ and $\mathbf{u}$. It follows that

$$\theta_{\mathbf{vu}} = \theta_{\mathbf{v}} - \theta_{\mathbf{u}}$$

$$\begin{aligned} \cos \theta_{\mathbf{vu}} &= \cos(\theta_{\mathbf{v}} - \theta_{\mathbf{u}}) \\ &= \cos(\theta_{\mathbf{v}}) \cos(\theta_{\mathbf{u}}) + \sin(\theta_{\mathbf{v}}) \sin(\theta_{\mathbf{u}}) \\ &= \frac{v_x}{\text{norm}(\mathbf{v})} \cdot \frac{u_x}{\text{norm}(\mathbf{u})} + \frac{v_y}{\text{norm}(\mathbf{v})} \cdot \frac{u_y}{\text{norm}(\mathbf{u})} \\ &= \frac{v_x u_x + v_y u_y}{\text{norm}(\mathbf{v}) \text{norm}(\mathbf{u})} = \frac{v_x u_x + v_y u_y}{\|\mathbf{v}\| \times \|\mathbf{u}\|} \end{aligned}$$

Dot Product (Review)

1. Recall that the dot product of two vectors as

$$\mathbf{v} \cdot \mathbf{u} = v_x u_x + v_y u_y$$

2. In short, the dot product is a scalar that results from multiplying corresponding components of the vectors and summing them up.
3. Using the dot product, we can rewrite the norm as

$$\text{norm}(\mathbf{v}) = \|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}}$$

$$\text{norm}(\mathbf{u}) = \|\mathbf{u}\| = \sqrt{\mathbf{u} \cdot \mathbf{u}}$$

Orthogonal (Perpendicular) Vectors

1. Using the dot product, we can rewrite

$$\cos \theta_{\mathbf{v}\mathbf{u}} = \frac{v_x u_x + v_y u_y}{\text{norm}(\mathbf{v})\text{norm}(\mathbf{u})} = \frac{\mathbf{v} \cdot \mathbf{u}}{\sqrt{\mathbf{v} \cdot \mathbf{v}}\sqrt{\mathbf{u} \cdot \mathbf{u}}}$$

$$\mathbf{v} \cdot \mathbf{u} = \cos \theta_{\mathbf{v}\mathbf{u}} \times \|\mathbf{v}\| \times \|\mathbf{u}\|$$

2. Recall two facts: (1) two lines are perpendicular if the angle between them is 90 degrees; (2) $\cos(90^\circ) = 0$.
3. It follows that

$$\cos \theta_{\mathbf{v}\mathbf{u}} = 0 \leftrightarrow \mathbf{v} \text{ and } \mathbf{u} \text{ are perpendicular} \leftrightarrow \mathbf{v} \cdot \mathbf{u} = 0$$

4. In short, two vectors are orthogonal if their dot product is zero (the angle is 90 degrees).

Applications of Vectors in Statistics

- **Principal Component Analysis (PCA):** PCA is used to reduce the dimensionality of large datasets by finding a set of orthogonal vectors (principal components) that capture the most variance.
- **Linear Regression:** The coefficients in a linear regression model can be viewed as vector operations, where the goal is to find the vector (of coefficients) that minimizes the distance (error) between the predicted and actual data points.
- **Correlation and Covariance:** These can be understood as dot products between standardized vectors, representing the relationship between different variables.
- **Distance Metrics:** Euclidean distance, used in clustering and classification, is the norm of the difference vector between two points in space.

OLS produces orthogonal decomposition

1. Consider the simple regression $y_i = \beta_1 x_i + e_i$. Recall the first-order condition of minimizing the $RSS(\beta_1) = \sum e_i^2 = \sum (y_i - \beta_1 x_i)^2$:

$$\frac{dRSS}{d\beta_1} = \sum -2x_i(y_i - \hat{\beta}_1 x_i) = 0$$
$$\sum x_i \hat{e}_i = 0$$

2. The left-hand side is just the inner product of $x = (x_1, x_2, \dots, x_n)$ and $\hat{e} = (\hat{e}_1, \hat{e}_2, \dots, \hat{e}_n)$. So an equivalent way to say FOC is that x and residual are orthogonal
3. In short, OLS decomposes y into two parts that are perpendicular (Exercise: Show that $\hat{y}$ and the residual $\hat{e}$ are orthogonal)

$$y = \hat{y} + \hat{e} \quad (\text{orthogonal OLS decomposition})$$

Using the Dot Product to Compute Sample Correlation

Definition of Sample Correlation:

$$r_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}}$$

Vector Notation:

$$\mathbf{x} = (x_1, x_2, \dots, x_n), \quad \mathbf{y} = (y_1, y_2, \dots, y_n)$$

Centered Vectors:

$$\mathbf{x}_c = \mathbf{x} - \bar{x}, \quad \mathbf{y}_c = \mathbf{y} - \bar{y}$$

Dot Product Approach:

$$r_{xy} = \frac{\mathbf{x}_c \cdot \mathbf{y}_c}{\|\mathbf{x}_c\| \|\mathbf{y}_c\|}$$

Conclusion: *The correlation is the cosine of the angle between the centered data vectors.*

Basis, Span, and Space

Expressing Vectors as Linear Combinations of Basis Vectors

- Any vector in a vector space can be expressed as a linear combination of basis vectors.
- In $\mathbb{R}^2$, the standard basis vectors are:

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

- Any vector $\mathbf{v} = \begin{pmatrix} v_x \\ v_y \end{pmatrix}$ in $\mathbb{R}^2$ can be expressed as:

$$\mathbf{v} = v_x \mathbf{e}_1 + v_y \mathbf{e}_2 = v_x \begin{pmatrix} 1 \\ 0 \end{pmatrix} + v_y \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

- This concept is foundational for understanding vector spaces and the representation of vectors in different coordinate systems.

Expressing Vectors as Linear Combinations of Basis Vectors

- In $\mathbb{R}^3$, the standard basis vectors are:

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{e}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

- Any vector $\mathbf{v} = \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix}$ in $\mathbb{R}^3$ can be expressed as:

$$\mathbf{v} = v_x \mathbf{e}_1 + v_y \mathbf{e}_2 + v_z \mathbf{e}_3 = v_x \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + v_y \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + v_z \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Importance of Basis Vectors

- Basis vectors provide a framework for describing every vector in a vector space.
- They allow us to uniquely represent vectors using a set of coefficients (coordinates).
- The choice of basis determines the coordinate system and affects how we perform computations, like transformations and projections.
- In $\mathbb{R}^n$, any vector can be expressed as a linear combination of basis vectors.
- The number of basis vectors corresponds to the dimension of the vector space.

Span of a Linear Combination of Vectors

- The **span of a set of vectors** is the set of all possible linear combinations of those vectors.
- For vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$, the span is defined as:

$$\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\} = \{c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_n\mathbf{v}_n \mid c_1, c_2, \dots, c_n \in \mathbb{R}\}$$

- Geometrically:
 - In $\mathbb{R}^2$, the span of two linearly independent vectors forms a plane.
 - In $\mathbb{R}^3$, the span of two linearly independent vectors forms a plane, and the span of three linearly independent vectors fills the 3D space.
- The concept of span is fundamental for understanding vector spaces, subspaces, and the solutions to systems of linear equations.

Vector Space

- A **vector space** (or linear space) is a set of vectors that satisfies two main properties:
 - **Vector Addition**: If $\mathbf{u}$ and $\mathbf{v}$ are vectors in the space, then their sum $\mathbf{u} + \mathbf{v}$ is also in the space.
 - **Scalar Multiplication**: If $\mathbf{v}$ is a vector in the space and c is a scalar, then $c\mathbf{v}$ is also in the space.
- Examples of vector spaces include:
 - $\mathbb{R}^2$: The set of all 2-dimensional vectors.
 - $\mathbb{R}^3$: The set of all 3-dimensional vectors.
- A vector space must also include the zero vector (the vector where all components are zero) and must be closed under vector addition and scalar multiplication.

Subspace

- A **subspace** is a subset of a vector space that is itself a vector space under the same operations of addition and scalar multiplication.
- To be a subspace, a subset S of a vector space V must satisfy three conditions:
 - **Non-empty**: The subspace must contain the zero vector.
 - **Closed under addition**: If $\mathbf{u}$ and $\mathbf{v}$ are in S , then $\mathbf{u} + \mathbf{v}$ must also be in S .
 - **Closed under scalar multiplication**: If $\mathbf{v}$ is in S and c is a scalar, then $c\mathbf{v}$ must also be in S .
- Intuitively, a vector space is like the entire space that contains all possible vectors, while a subspace is a smaller “slice” or subset of that space that still retains the same structural properties as the larger space.

Linear Independence

Intuition Behind Linear Independence

- Imagine vectors as arrows in space; some arrows can be expressed as a combination of others.
- If a vector can be written as a combination of others, it doesn't add new direction to the space.
- Vectors that provide unique directions (and cannot be expressed as combinations of others) are said to be linearly independent.
- Linear independence means that each vector contributes something “new” to the span, expanding the space.

Visualizing Linear Independence in $\mathbb{R}^2$

- Consider two vectors in $\mathbb{R}^2$:
 - If they point in the same direction (are collinear), they are not linearly independent.
 - If they point in different directions, they are linearly independent and span the entire plane.
- **Example:**
 - Vectors $\mathbf{u} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ are linearly independent.
 - They span the entire $\mathbb{R}^2$ space.

Visualizing Linear Independence in $\mathbb{R}^3$

- Consider three vectors in $\mathbb{R}^3$:
 - If all three lie in the same plane, they are not linearly independent.
 - If they each point in different, non-coplanar directions, they are linearly independent and span the entire 3D space.
- **Example:**
 - Vectors $\mathbf{u} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $\mathbf{v} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, and $\mathbf{w} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ are linearly independent.
 - They span the entire $\mathbb{R}^3$ space.

Formal Definition of Linear Independence

- A set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ in a vector space is **linearly independent** if the only solution to the equation

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_n\mathbf{v}_n = \mathbf{0}$$

is $c_1 = c_2 = \dots = c_n = 0$.

- If there exists a non-trivial solution (where some coefficients c_i are not zero), the vectors are **linearly dependent**.
- Linear independence implies that no vector in the set can be expressed as a linear combination of the others.

Checking Linear Independence

- **Problem:** Determine whether the vectors $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$, $\mathbf{v}_2 = \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$,

and $\mathbf{v}_3 = \begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix}$ in $\mathbb{R}^3$ are linearly independent.

1. Set up the linear combination equation:

$$\mathbf{0} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 = c_1 \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + c_2 \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix} + c_3 \begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

2. This leads to the system of linear equations:

$$c_1 + 4c_2 + 7c_3 = 0$$

$$2c_1 + 5c_2 + 8c_3 = 0$$

$$3c_1 + 6c_2 + 9c_3 = 0$$

3. Check whether the system has a unique solution $c_1 = c_2 = c_3 = 0$

Why Set the Linear Combination to Zero?

- The key question is:
 - “Is there any non-trivial way (where not all coefficients are zero) to combine these vectors to get nothing (the zero vector)?”
- **If the answer is “no”:**
 - The only solution is the trivial one where all coefficients are zero.
 - This means the vectors must be independent, as no vector can be written as a combination of the others.
- **If the answer is “yes”:**
 - There is some non-trivial combination that results in the zero vector.
 - This implies that at least one vector is redundant—it can be built from the others, meaning the vectors are linearly dependent.

Systems of Equations

Possible Solutions for a System of Linear Equations

- Consider the system of linear equations:

$$c_1 + 4c_2 + 7c_3 = 0$$

$$2c_1 + 5c_2 + 8c_3 = 0$$

$$3c_1 + 6c_2 + 9c_3 = 0$$

- This system can have one of three outcomes:**
 - One Unique Solution:** Occurs if the system is consistent and has exactly as many independent equations as unknowns.
 - No Solution:** Occurs if the system is inconsistent (e.g., parallel planes that never intersect).
 - Infinite Solutions:** Occurs if the system has fewer independent equations than unknowns, leading to free variables.
- Why This Holds for Any System:**
 - A system of linear equations represents geometric objects (lines, planes, etc.).
 - The nature of intersections of these objects (points, lines, entire planes) dictates the number of solutions.

What is a Matrix?

- A **matrix** A is a rectangular array of numbers arranged in rows and columns.
- Matrices are used to represent and solve systems of linear equations compactly.
- Matrices also generalize the concept of linear function for vectors.
- The given system of equations:

$$c_1 + 4c_2 + 7c_3 = 0$$

$$2c_1 + 5c_2 + 8c_3 = 0$$

$$3c_1 + 6c_2 + 9c_3 = 0$$

- Can be expressed as:

$$A\mathbf{c} = \begin{pmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Linear Systems: From Equations to Matrix Form

More generally:

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2$$

$$\vdots = \vdots$$

$$a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n = b_n$$

Which can be expressed as:

$$\mathbf{Ax} = \mathbf{b}$$

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix}$$

Matrix Multiplication Procedure

- **Matrix Multiplication:**

- To multiply an $n \times n$ matrix A by an $n \times k$ matrix B , each element c_{ij} of the resulting $n \times k$ matrix C is computed as:

$$c_{ij} = \sum_{r=1}^n a_{ir} \cdot b_{rj}$$

- In other words, each element of the result is the dot product of the corresponding row of the first matrix and the column of the second matrix.

- **Conformity of Matrix Multiplication:**

- The multiplication is only defined when the inner dimensions match, i.e., the number of columns in A must be the same as the number of rows in B .
- If A is an $m \times n$ matrix and B is an $n \times k$ matrix, the product $A \times B$ will result in an $m \times k$ matrix.
- In general, $A \times B \neq B \times A$.

Matrix Multiplication Procedure

- **Example:**

- Multiply a 2×2 matrix by a 2×1 vector:

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} b_{11} \\ b_{21} \end{pmatrix} = \begin{pmatrix} a_{11}b_{11} + a_{12}b_{21} \\ a_{21}b_{11} + a_{22}b_{21} \end{pmatrix}$$

- Example with numbers:

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 \\ 6 \end{pmatrix} = \begin{pmatrix} 1(5) + 2(6) = 17 \\ 3(5) + 4(6) = 39 \end{pmatrix}$$

Matrix Multiplication Procedure: Another Example

Example:

Multiply a 3×2 matrix by a 2×3 matrix:

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{pmatrix} = \begin{pmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} & a_{11}b_{13} + a_{12}b_{23} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} & a_{21}b_{13} + a_{22}b_{23} \\ a_{31}b_{11} + a_{32}b_{21} & a_{31}b_{12} + a_{32}b_{22} & a_{31}b_{13} + a_{32}b_{23} \end{pmatrix}$$

Example with numbers:

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix} \begin{pmatrix} 7 & 8 & 9 \\ 10 & 11 & 12 \end{pmatrix} = \begin{pmatrix} 1(7) + 2(10) = 27 & 1(8) + 2(11) = 30 & 1(9) + 2(12) = 33 \\ 3(7) + 4(10) = 61 & 3(8) + 4(11) = 68 & 3(9) + 4(12) = 75 \\ 5(7) + 6(10) = 95 & 5(8) + 6(11) = 106 & 5(9) + 6(12) = 117 \end{pmatrix}$$

Row Operations and Linear Systems

- **Row Operations:**

- Row operations are procedures that can be performed on the rows of a matrix without changing the solution set of the corresponding linear system.
- There are three types of row operations:
 1. **Row Swapping:** Interchange two rows.
 2. **Row Multiplication:** Multiply all elements of a row by a non-zero scalar.
 3. **Row Addition:** Add a multiple of one row to another row.

- **Relation to Linear Systems:**

- Each row of a matrix in a linear system corresponds to an equation.
- Row operations are equivalent to performing the same operations on the corresponding equations in the system.
- These operations are used to simplify a system of equations, typically with the goal of solving the system by transforming the matrix into a simpler form, such as row-echelon form or reduced row-echelon form.

Transforming to Row Echelon Form

- Recall the system of equations to check if the vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are linearly independent:
- Given System of Equations:**

$$c_1 + 4c_2 + 7c_3 = 0$$

$$2c_1 + 5c_2 + 8c_3 = 0$$

$$3c_1 + 6c_2 + 9c_3 = 0$$

- Corresponding Augmented Matrix:**

$$\left(\begin{array}{ccc|c} 1 & 4 & 7 & 0 \\ 2 & 5 & 8 & 0 \\ 3 & 6 & 9 & 0 \end{array} \right)$$

Transforming to Row Echelon Form

- **Step 1: Eliminate the first column below the pivot (1st row):**
 - Subtract 2 times the first row from the second row:

$$R_2 \rightarrow R_2 - 2R_1$$

- Subtract 3 times the first row from the third row:

$$R_3 \rightarrow R_3 - 3R_1$$

$$\left(\begin{array}{ccc|c} 1 & 4 & 7 & 0 \\ 0 & -3 & -6 & 0 \\ 0 & -6 & -12 & 0 \end{array} \right)$$

Transforming to Row Echelon Form

- **Step 2: Eliminate the second column below the pivot (2nd row):**
 - Subtract 2 times the second row from the third row:

$$R_3 \rightarrow R_3 - 2R_2$$

$$\left(\begin{array}{ccc|c} 1 & 4 & 7 & 0 \\ 0 & -3 & -6 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

- **Resulting Row Echelon Form:**
 - The matrix is now in row echelon form, where all the elements below the leading coefficients (pivots) are zeros.
 - Note: A *Reduced* Row Echelon Form matrix as the pivots equal to 1.

Row of Zeros and Parametrization of Solutions

- **Row of Zeros:**

- After transforming the matrix to row echelon form, we obtained a row of zeros:

$$\left(\begin{array}{ccc|c} 1 & 4 & 7 & 0 \\ 0 & -3 & -6 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

- This indicates that the system of equations has fewer independent equations than variables.
 - The presence of a row of zeros suggests that the system does not have a unique solution, but rather an infinite number of solutions.
- Importantly, we learned that the vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are **linearly dependent** (i.e., not linearly independent).

Implication for the Span of Linearly Dependent Vectors

- **Linearly Dependent Vectors:**
 - A set of vectors is linearly dependent if at least one vector in the set can be expressed as a linear combination of the others.
- **Implication for the Span:**
 - The span of a set of linearly dependent vectors is the same as the span of a smaller subset of those vectors.
 - This means that the linearly dependent vectors do not contribute additional dimensions to the span.
 - In other words, removing the dependent vectors does not change the span; it remains the same subspace.
- When vectors are linearly dependent, the set does not span the full n -dimensional space. Instead, it spans a lower-dimensional subspace.

Row of Zeros and Parametrization of Solutions

- **Parametrization of Solutions:**

- With fewer independent equations, one or more variables can be treated as free parameters.
- These free variables can take any value, and the remaining variables can be expressed in terms of these free variables.
- For the current system, if we let $c_3 = t$ (where t is a free parameter), the system can be solved for c_1 and c_2 in terms of t .
- This approach gives us a parametric description of the solution set, representing an infinite family of solutions.

Parametrized Solution

- **System in Row Echelon Form:**

$$\left(\begin{array}{ccc|c} 1 & 4 & 7 & 0 \\ 0 & -3 & -6 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

- **Parametrization:**

$$c_3 = t \quad (\text{free parameter})$$

$$c_2 = 2t$$

$$c_1 = -10t$$

- **Solution:**

$$\begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = t \begin{pmatrix} -10 \\ 2 \\ 1 \end{pmatrix} \quad \text{where } t \in \mathbb{R}$$

Rank of a Matrix and Solutions of Linear Systems

- **Rank of a Matrix:**

- The **rank** of a matrix is the maximum number of linearly independent rows (or columns) in the matrix.
- It represents the row space or column space dimension, which is the number of pivots in row echelon form. In other words, it is the dimension of the vector space spanned by its rows or columns.

- **Theorem (Rank and Solutions):** For a system of linear equations $A\mathbf{x} = \mathbf{b}$:

- If $\text{rank}(A) = \text{rank}([A|\mathbf{b}])$ and $\text{rank}(A) = n$, the system has a unique solution (the matrix A has full rank).
- If $\text{rank}(A) = \text{rank}([A|\mathbf{b}]) < n$, the system has infinitely many solutions.
- If $\text{rank}(A) \neq \text{rank}([A|\mathbf{b}])$, the system has no solution.

Matrices as Linear Transformations

- **Linear Transformation:**

- A linear transformation is a function $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ that maps vectors from one space to another while preserving vector addition and scalar multiplication.
- Mathematically, $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ and $T(c\mathbf{u}) = cT(\mathbf{u})$.

- **Matrices as Representations of Linear Transformations:**

- Every linear transformation can be represented by a matrix A such that for a vector $\mathbf{x}$ in $\mathbb{R}^n$, the transformation is given by:

$$T(\mathbf{x}) = A\mathbf{x}$$

- The matrix A encodes how the transformation acts on the basis vectors of the space.

- **Geometric Interpretation:**

- Matrices can stretch, compress, rotate, reflect, or shear vectors in a space, depending on the entries of the matrix.

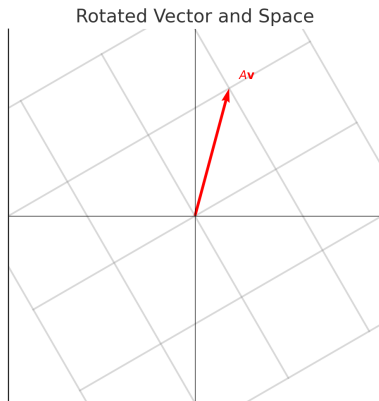
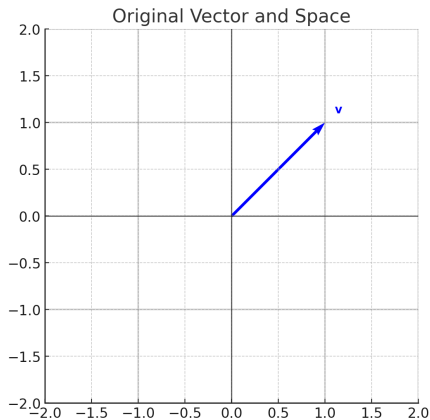
Matrices and Changes of Basis

- **Matrix Representation of Basis Vectors:**
 - A matrix where each column corresponds to a basis vector can represent a linear transformation.
 - This matrix can be used to transform vectors from one space (or basis) to another.
- **Linear Transformation as Change of Basis:**
 - The matrix effectively transforms vectors expressed in one basis into vectors expressed in another.
 - Thus, linear transformations can be viewed as changes of basis, altering how we represent vectors in the space.
- **Key Condition: Linear Independence:**
 - The basis vectors (columns of the matrix) must be linearly independent to ensure the transformation is valid and invertible.

Rotation Matrix and Its Effect on Space

Rotation Matrix A (Angle θ)

$$A = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$



Matrix Multiplication as Composite Linear Transformations

- **Matrix Multiplication:**

- Given two $n \times n$ matrices A and B , their product $C = AB$ represents the result of applying two linear transformations in sequence.

- **Composite Function of Transformations:**

- Think of each matrix as a linear transformation.
- First, apply transformation B to a vector $\mathbf{v}$: $\mathbf{v}' = B\mathbf{v}$.
- Then, apply transformation A to the result: $\mathbf{v}'' = A\mathbf{v}' = AB\mathbf{v}$.

- **Conclusion:**

- Multiplying matrices A and B is equivalent to performing the linear transformation B followed by A , akin to a composite function.

Example of Matrix Multiplication

$$X = \begin{pmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{pmatrix}, \quad Y = \begin{pmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{pmatrix}$$

$$XY = \begin{pmatrix} x_{11}y_{11} + x_{12}y_{21} & x_{11}y_{12} + x_{12}y_{22} \\ x_{21}y_{11} + x_{22}y_{21} & x_{21}y_{12} + x_{22}y_{22} \end{pmatrix}$$

$$\begin{aligned} & \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \times \begin{pmatrix} 5 & 6 & 7 \\ 8 & 9 & 10 \end{pmatrix} \\ &= \begin{pmatrix} (1) \times 5 + (2) \times 8 & (1) \times 6 + (2) \times 9 & (1) \times 7 + (2) \times 10 \\ (3) \times 5 + (4) \times 8 & (3) \times 6 + (4) \times 9 & (3) \times 7 + (4) \times 10 \end{pmatrix} \\ &= \begin{pmatrix} 21 & 24 & 27 \\ 47 & 54 & 61 \end{pmatrix} \end{aligned}$$

Intuitive Understanding of the Inverse of a Square Matrix

- **What is the Inverse of a Matrix?**

- The inverse of a square matrix $A_{n \times n}$, denoted by A^{-1} , is another matrix that “undoes” the transformation applied by A .
- Mathematically, $AA^{-1} = A^{-1}A = I$, where I is the identity matrix.

- **Intuitive Explanation:**

- Think of a matrix as a function that transforms vectors in a space. The inverse matrix reverses this transformation, bringing vectors back to their original positions.
- *Example:* If A rotates and scales a vector, A^{-1} rotates and scales it back in the opposite direction, returning it to its original state.

- **Key Points:**

- Not all matrices have inverses. A matrix has an inverse only if it is **non-singular** (i.e., it has full rank, and its determinant is non-zero).
- The inverse matrix is unique—each invertible matrix has exactly one inverse.

Inverse of a 2×2 Matrix and Determinant

- **Generic 2×2 Matrix:**

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

- **Determinant of A :**

$$\det(A) = ad - bc$$

- The determinant is a scalar that determines whether A is invertible.
- If $\det(A) = 0$, A is singular and has no inverse.

- **Inverse of A (when $\det(A) \neq 0$):**

$$A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

- The inverse exists only if $\det(A) \neq 0$.
- A^{-1} “undoes” the transformation applied by A .

Row Operations as Linear Transformations

Theorem: Any row operation on a matrix A can be represented as multiplication by an elementary matrix E . If E corresponds to the row operation, then:

$$\text{Row Operation on } A \equiv E \cdot A$$

Example: Given $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, perform the row operation “Add -3 times the first row to the second row”. The corresponding elementary matrix is :

$$E = \begin{pmatrix} 1 & 0 \\ -3 & 1 \end{pmatrix}$$

Apply the transformation:

$$E \cdot A = \begin{pmatrix} 1 & 0 \\ -3 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 1 \cdot 1 + 0 \cdot 3 & 1 \cdot 2 + 0 \cdot 4 \\ (-3 \cdot 1) + 1 \cdot 3 & (-3 \cdot 2) + 1 \cdot 4 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 0 & -2 \end{pmatrix}$$

The Effect of Row Operations on Determinants

- **Theorem. Row Operations and Determinants:**
 - **Row Addition/Subtraction:** Adding or subtracting a multiple of one row to/from another does not change the determinant.
 - **Row Swapping:** Swapping two rows changes the sign of the determinant.
 - **Row Scaling:** Multiplying a row by a scalar k scales the determinant by k .
- **Key Insight:**
 - If you apply only row addition/subtraction operations, the determinant remains unchanged regardless of how many such operations you perform.
 - **Caution:** If your sequence of operations includes row swaps or row scalings, the determinant will change according to the rules above.

Upper Triangular Matrix and Its Determinant

- **Upper Triangular Matrix:**

- An $n \times n$ matrix U is called an **upper triangular matrix** if all the elements below the main diagonal are zero. Formally,

$$U = \begin{pmatrix} u_{11} & u_{12} & \cdots & u_{1n} \\ 0 & u_{22} & \cdots & u_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & u_{nn} \end{pmatrix}$$

- **Determinant of an Upper Triangular Matrix:**

- **Formula:** For U , the determinant is given by:

$$\det(U) = u_{11} \cdot u_{22} \cdot \dots \cdot u_{nn} = \prod_{i=1}^n u_{ii}$$

Upper Triangular Block Matrix and Its Determinant

- **Upper Triangular Block Matrix:**

- A matrix is called an **upper triangular block matrix** if it consists of square submatrices on the diagonal and zero submatrices below the diagonal. Formally, for a block matrix U :

$$U = \begin{pmatrix} A_{11} & A_{12} & \cdots & A_{1m} \\ 0 & A_{22} & \cdots & A_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & A_{mm} \end{pmatrix}$$

where A_{ii} are square submatrices and all blocks below the diagonal are zero matrices.

- **Determinant of an Upper Triangular Block Matrix:**

- **Formula:** The determinant of an upper triangular block matrix is the product of the determinants of the diagonal blocks:

$$\det(U) = \det(A_{11}) \cdot \det(A_{22}) \cdot \dots \cdot \det(A_{mm}) = \prod_{i=1}^m \det(A_{ii})$$

Determinants, Elementary Matrices, and Row Echelon Form

- **Lemma 1: Determinant of an Elementary Matrix**

- The determinant of any elementary matrix E is always non-zero.
- **Implication:** Since elementary matrices represent row operations, which are invertible transformations, their determinants are non-zero, preserving the structure of the matrix.

- **Lemma 2: Determinant and Row Echelon Form**

- Given that $A_{\text{REF}} = E_n E_{n-1} \cdots E_1 A$, the determinant of a matrix in row echelon form A_{REF} is zero if and only if $\det(A) = 0$.
- **Key Insight:**
If $\det(A) = 0$, then during the row reduction process to A_{REF} , at least one row will consist entirely of zeros, indicating that the matrix is singular.

Determinant and the Number of Solutions

- **Determinant as a Criterion:**

- The determinant of a matrix A gives crucial insight into the number of solutions of a linear system $A\mathbf{x} = \mathbf{b}$.
- **Key Insight:** If $\det(A) = 0$, the matrix is singular, indicating that the system may have either no solutions or infinitely many solutions.

- **Row Echelon Form and Zero Rows:**

- When $\det(A) = 0$, row operations transform A into a row echelon form that has at least one row of zeros.
- **Implication:** A zero row in the row echelon form signifies that the rank of A is less than its dimension, impacting the uniqueness of solutions.

Matrix Operations

Matrix Operations as Compositions of Linear Transformations

- **Matrix Addition (Element-Wise Definition):**

$$C = A + B$$

$$c_{ij} = a_{ij} + b_{ij} \quad \text{for all } i, j$$

- **Matrix Subtraction (Element-Wise Definition):**

$$D = A - B$$

$$d_{ij} = a_{ij} - b_{ij} \quad \text{for all } i, j$$

- **Multiplication by a Constant (Element-Wise Definition):**

$$E = k \times A$$

$$e_{ij} = k \times a_{ij} \quad \text{for all } i, j$$

Matrix Transpose and Its Geometric Interpretation

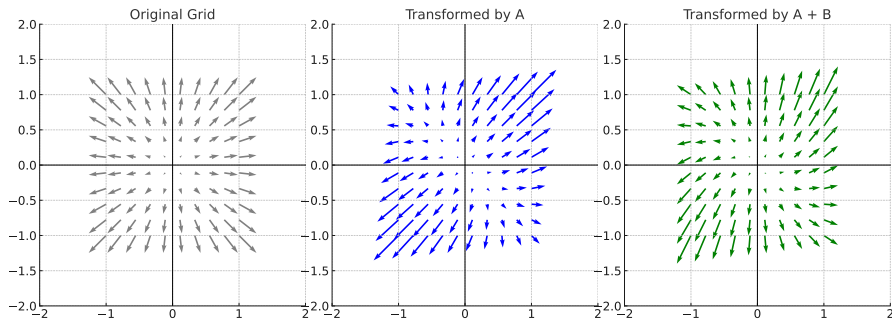
Matrix Transpose:

- The transpose of a matrix A , denoted A^T or A' , is obtained by swapping its rows and columns.
- Element-wise: $(a^T)_{ij} = a_{ji}$.

Geometric Interpretation:

- As a linear transformation, transposing a matrix reflects vectors across the line $y = x$ in $\mathbb{R}^2$.
- In higher dimensions, it reflects vectors across the hyperplane where the corresponding coordinates are equal.

$A + B$ as a Composition of Linear Transformations



$$A = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix} \quad B = \begin{pmatrix} 0.5 & -0.5 \\ 0.5 & 1 \end{pmatrix}$$

Properties of Matrix Addition

1. $A + 0 = A - 0 = A$
2. $A + B = B + A$ (Commutativity of Matrix Addition)
3. $(A + B) + C = A + (B + C)$ (Associativity of Matrix Addition)
4. $\lambda(A + B) = \lambda A + \lambda B$ (Distributive Law for Matrix Addition)
5. $(\lambda + \mu)A = \lambda A + \mu A$
6. $(A + B)' = A' + B'$

Using Element-Wise Definitions Prove: $(A + B)' = A' + B'$

Properties of Matrix Multiplication

- (i) $(AB)C = A(BC)$ (*Associativity of Matrix Multiplication*)
- (ii) $AI = IA = A$
- (iii) $(A + B)C = AC + BC$ (*Distrib. Law for Matrix Mult.*)
- (iv) $A(\lambda B) = \lambda(AB)$ (*Distrib. Law for Scalar Multipl.*)
- (v) $(AB)^T = B^T A^T$ (*Transpose of a Product*)
and $(A_1 A_2 \dots A_k)^T = A_k^T \dots A_2^T A_1^T$

Some Special Types of Matrices

- **A square matrix** is a matrix that has an equal number of columns and rows.
- **A zero matrix** is a matrix in which all elements are 0.
- **A diagonal matrix** is a square matrix in which all elements other than those on the main diagonal are zero.
- **An identity matrix**, $I_{n \times n}$, is a diagonal matrix in which all elements on the main diagonal are 1. $AI = IA = A$
- **A symmetric matrix** is a square matrix in which the elements are symmetric about the main diagonal, or more formally one in which $a_{ij} = a_{ji}$.
 - If A is symmetric, $A = A^T$.
- **A lower (upper) triangular matrix** has non-zero elements only on or below (above) the main diagonal.
- **A submatrix** of an element is the matrix that remains when we take out the row and column in which the element is (so it has one fewer column and row than the original).

Trace

The trace of an $n \times n$ square matrix is the sum of its diagonal elements.

$$\text{tr}(A_{n \times n}) = a_{11} + a_{22} + \cdots + a_{nn} = \sum_{i=1}^n a_{ii}$$

Properties:

1. $\text{tr}(A) = \text{tr}(A')$
2. $\text{tr}(kA) = k \cdot \text{tr}(A)$
3. $\text{tr}(A \pm B) = \text{tr}(A) \pm \text{tr}(B)$
4. $\text{tr}(I_n) = n$
5. $\text{tr}(AB) = \text{tr}(BA)$, if both AB and BA exist

Exercise: Prove 5

Properties of the Inverse Matrix

1. $(A^{-1})^{-1} = A$
2. Multiplicative property: $(AB)^{-1} = B^{-1}A^{-1}$
3. Scalar multiplication: $(cA)^{-1} = c^{-1}A^{-1}$ if $c \neq 0$
4. $(A')^{-1} = (A^{-1})'$ Prove it! (hint: $(A')(A')^{-1} = I$)

Invertibility of a Matrix A

For a square matrix $A_{n \times n}$, **The following statements are equivalent:**

1. A is a non-singular matrix.
2. A has full rank.
3. $\det(A) \neq 0$.
4. A^{-1} exists.
5. A has n linearly independent columns.

Solving the Matrix Equation $Ax = b$

Context: To solve the matrix equation $Ax = b$, we can multiply both sides by the inverse of A , if A is non-singular. The steps below show how this leads to the solution for x :

$$A^{-1}(Ax) = A^{-1}b$$

$$(A^{-1}A)x = A^{-1}b$$

$$I_n x = A^{-1}b$$

$$x = A^{-1}b$$

Intro to Multi- variate Calculus

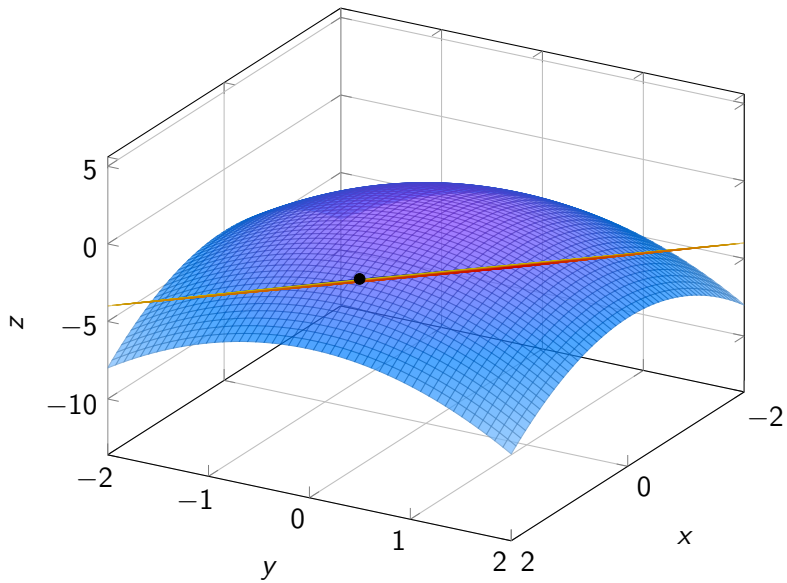
Multivariate Functions and the Gradient

Multivariate Function: A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ that takes multiple inputs $x_1, x_2, \dots, x_n$ and maps them to a single output $f(x_1, x_2, \dots, x_n)$.

Gradient: The gradient $\nabla f(x)$ of a multivariate function is a vector of partial derivatives:

$$\nabla f(x) = \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n} \right)$$

Interpretation: The gradient points in the direction of the steepest ascent of the function, and its magnitude indicates the rate of increase.



Gradient of a Multivariate Function

Consider a multivariate function $f : \mathbb{R}^n \rightarrow \mathbb{R}$. The gradient of f at a point $\mathbf{x} \in \mathbb{R}^n$ is defined as:

$$\nabla f(\mathbf{x}) = \begin{bmatrix} \frac{\partial f}{\partial x_1}(\mathbf{x}) \\ \frac{\partial f}{\partial x_2}(\mathbf{x}) \\ \vdots \\ \frac{\partial f}{\partial x_n}(\mathbf{x}) \end{bmatrix}$$

Interpretation:

- The gradient $\nabla f(\mathbf{x})$ is a vector that points in the direction of the steepest ascent of f at $\mathbf{x}$.
- Each component represents the rate of change of f with respect to each variable.

Second Derivatives and the Hessian Matrix

To understand how the function changes more comprehensively, we can consider the second derivatives, which form the Hessian matrix. The Hessian $H_f(\mathbf{x})$ of f at $\mathbf{x}$ is defined as:

$$H_f(\mathbf{x}) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2}(\mathbf{x}) & \frac{\partial^2 f}{\partial x_1 \partial x_2}(\mathbf{x}) & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n}(\mathbf{x}) \\ \frac{\partial^2 f}{\partial x_2 \partial x_1}(\mathbf{x}) & \frac{\partial^2 f}{\partial x_2^2}(\mathbf{x}) & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n}(\mathbf{x}) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1}(\mathbf{x}) & \frac{\partial^2 f}{\partial x_n \partial x_2}(\mathbf{x}) & \cdots & \frac{\partial^2 f}{\partial x_n^2}(\mathbf{x}) \end{bmatrix}$$

Key Points:

- The Hessian matrix $H_f(\mathbf{x})$ is symmetric.
- The diagonal elements $\frac{\partial^2 f}{\partial x_i^2}$ represent the curvature of f along each coordinate axis.
- The off-diagonal elements $\frac{\partial^2 f}{\partial x_i \partial x_j}$ represent how the variables interact.

Deriving the Hessian from the Gradient

The Hessian can be derived by taking the derivative of the gradient vector. Given:

$$\nabla f(\mathbf{x}) = \begin{bmatrix} \frac{\partial f(\mathbf{x})}{\partial x_1} \\ \frac{\partial f(\mathbf{x})}{\partial x_2} \\ \vdots \\ \frac{\partial f(\mathbf{x})}{\partial x_n} \end{bmatrix},$$

the Hessian is obtained by differentiating each component of the gradient with respect to each variable:

$$H_f(\mathbf{x}) = \begin{bmatrix} \frac{\partial}{\partial x_1} \left(\frac{\partial f}{\partial x_1} \right) & \frac{\partial}{\partial x_2} \left(\frac{\partial f}{\partial x_1} \right) & \cdots & \frac{\partial}{\partial x_n} \left(\frac{\partial f}{\partial x_1} \right) \\ \frac{\partial}{\partial x_1} \left(\frac{\partial f}{\partial x_2} \right) & \frac{\partial}{\partial x_2} \left(\frac{\partial f}{\partial x_2} \right) & \cdots & \frac{\partial}{\partial x_n} \left(\frac{\partial f}{\partial x_2} \right) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial}{\partial x_1} \left(\frac{\partial f}{\partial x_n} \right) & \frac{\partial}{\partial x_2} \left(\frac{\partial f}{\partial x_n} \right) & \cdots & \frac{\partial}{\partial x_n} \left(\frac{\partial f}{\partial x_n} \right) \end{bmatrix}$$

Taylor Approximation in Vector Form

First-order Approximation

Given a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the first-order Taylor approximation at point $\mathbf{x}$ is:

$$f(\mathbf{x} + \mathbf{h}) \approx f(\mathbf{x}) + \nabla f(\mathbf{x})^T \mathbf{h}$$

Second-order Approximation

The second-order Taylor approximation incorporates the Hessian:

$$f(\mathbf{x} + \mathbf{h}) \approx f(\mathbf{x}) + \nabla f(\mathbf{x})^T \mathbf{h} + \frac{1}{2} \mathbf{h}^T H_f(\mathbf{x}) \mathbf{h}$$

- $\nabla f(\mathbf{x})$: Gradient of f at $\mathbf{x}$.
- $H_f(\mathbf{x})$: Hessian matrix of f at $\mathbf{x}$.

Conceptual View

Conceptually:

- The gradient $\nabla f(\mathbf{x})$ gives the direction of steepest ascent.
- The Hessian $H_f(\mathbf{x})$ gives a quadratic approximation of how f changes, describing the curvature around $\mathbf{x}$.
- The Hessian is particularly important in optimization problems to determine the nature (concave, convex, or saddle point) of critical points.

Linear Regression as a Geometrical Problem - Interpretation

- Each observation y_i can be represented as a point in $\mathbb{R}^{p+1}$, corresponding to the vector:

$$y_i = \beta_0 + \beta \cdot \mathbf{x}_i + \epsilon_i$$

where $\mathbf{x}_i = \begin{pmatrix} x_{i1} \\ x_{i2} \\ \vdots \\ x_{ip} \end{pmatrix}$, and $\beta = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{pmatrix}$.

- The goal of linear regression is to find the vector $\hat{\beta}$ such that the predicted values $\hat{y}_i = \beta_0 + \hat{\beta} \cdot \mathbf{x}_i$ are as close as possible to the observed values y_i .
- Geometrically, this involves finding the orthogonal projection of each y_i onto the space spanned by the vectors $\mathbf{x}_i$ for all i .
- The error $\epsilon_i = y_i - \hat{y}_i$ represents the distance between the observed point and the predicted point on the regression line.

Vectorial and Summation Formulation of OLS

Model Equation

$$y_i = \beta_0 + \mathbf{x}_i^T \boldsymbol{\beta} + \epsilon_i = \beta_0 + \sum_{j=1}^p \beta_j x_{ij} + \epsilon_i$$

- Here, y_i represents the observed value for the i -th observation.
- $\mathbf{x}_i$ is the vector of predictors for the i -th observation, with $\mathbf{x}_i^T$ being its transpose.
- $\boldsymbol{\beta}$ is the vector of regression coefficients.
- ϵ_i is the error term for the i -th observation.

Objective Function to Minimize

$$J(\boldsymbol{\beta}) = \sum_{i=1}^n (y_i - \mathbf{x}_i^T \boldsymbol{\beta})^2$$

Gradients of Linear and Quadratic Forms

Linear Form:

$$f(\mathbf{x}) = \mathbf{a}^\top \mathbf{x}, \quad \nabla_{\mathbf{x}} f(\mathbf{x}) = \mathbf{a}.$$

Quadratic Form:

$$g(\mathbf{x}) = \mathbf{x}^\top A \mathbf{x}, \quad \nabla_{\mathbf{x}} g(\mathbf{x}) = 2A\mathbf{x}.$$

Key Points:

- The gradient of the linear form is the vector used in the linear combination.
- The gradient of the quadratic form is a linear transformation of the vector $\mathbf{x}$ by the matrix symmetric A , scaled by 2.

Minimization of Sum of Squared Residuals (SSR) - Part 1

Objective Function

The objective function for OLS, expressed in terms of index i , is:

$$J(\beta) = \sum_{i=1}^n (y_i - x_i^T \beta)^2$$

First-Order Condition (FOC)

To find the minimum of $J(\beta)$, compute the gradient with respect to β and set it to zero. For each coefficient β_j :

$$\frac{\partial J}{\partial \beta_j} = -2 \sum_{i=1}^n x_{ij} (y_i - x_i^T \beta) = 0$$

Minimization of Sum of Squared Residuals (SSR) - Part 2

First Order Condition

The FOC simplifies to the system of normal equations:

$$\sum_{i=1}^n x_{ij} y_i = \sum_{i=1}^n x_{ij} x_i^T \beta \quad \text{for each } j$$

- x_{ij} denotes the j -th element of the predictor vector x_i .
- x_i^T is the transpose of the predictor vector for the i -th observation.

Minimization of Sum of Squared Residuals (SSR) - Part 3

Matrix Formulation of Normal Equations

Define matrix X as:

$$X'X = \sum_{i=1}^n x_i x_i^T \quad X'Y = \sum_{i=1}^n x_i^T y_i$$

This represents the sum of the outer products of each predictor vector x_i with itself.

Inversion of Matrix $X'X$

Assuming $X'X$ is non-singular, invert $X'X$ to solve for β :

$$(X'X)^{-1} = \left(\sum_{i=1}^n x_i x_i^T \right)^{-1}$$

Minimization of Sum of Squared Residuals (SSR) - Part 4

Solution for β

Using the inverted matrix $(X^T X)^{-1}$, solve the normal equations to isolate β :

$$\beta = (X^T X)^{-1}(X^T y) = \left(\sum_{i=1}^n x_i x_i^T \right)^{-1} \left(\sum_{i=1}^n x_i y_i \right)$$

This equation gives the estimated coefficients by minimizing the SSR.